where the Ranger lands, temperatures are higher and the ice on which they land is presumably all frozen water.

Dr. Mann’s Geological Data

In the movie, Dr. Mann has been searching for organic material on his planet and he claims to have found promising evidence. Promising but not definitive. He shows his data to Brand and Romilly.

The data consist of field notes that indicate where each rock sample was collected and the geological environment there, together with chemical analyses of the sample. Those chemical analyses are Dr. Mann’s evidence of organics.

Figure 19.3 shows a page from these data. The data were actually prepared for the movie by Erika Swanson, a talented geology PhD student at Caltech. Erika has done fieldwork and chemical analyses somewhat similar to Dr. Mann’s.